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| 29 | Which of the following best describes the Heisenberg's uncertainty principle?                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                  |
|    | A                                                                                                                                                                                                                                                                                                                                                                                                                   | The uncertainty principle applies only to microscopic particles like electrons.                                                  |
|    | B                                                                                                                                                                                                                                                                                                                                                                                                                   | The uncertainty principle only applies to particles moving close to the speed of light.                                          |
|    | C                                                                                                                                                                                                                                                                                                                                                                                                                   | The product of the uncertainties in position and momentum of a particle has a minimum value.                                     |
|    | D                                                                                                                                                                                                                                                                                                                                                                                                                   | It is possible to reduce the uncertainty in both position and momentum of a particle to zero with advanced measuring techniques. |
|    | <p><b>Answer: C</b></p> <p><b>Explanation/Working:</b></p> <p>The HUP applies to all particles, not just electrons (Option A) or objects moving at high speeds (Option B). It is also a limitation of knowledge of the value of the quantities, and not just a technological issue (Option D).</p> <p>Option C is correct because this is the statement of the inequality <math>\Delta x \Delta p \geq h</math></p> |                                                                                                                                  |

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| 20 | Thorium 232 undergoes the following fission reaction: |
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